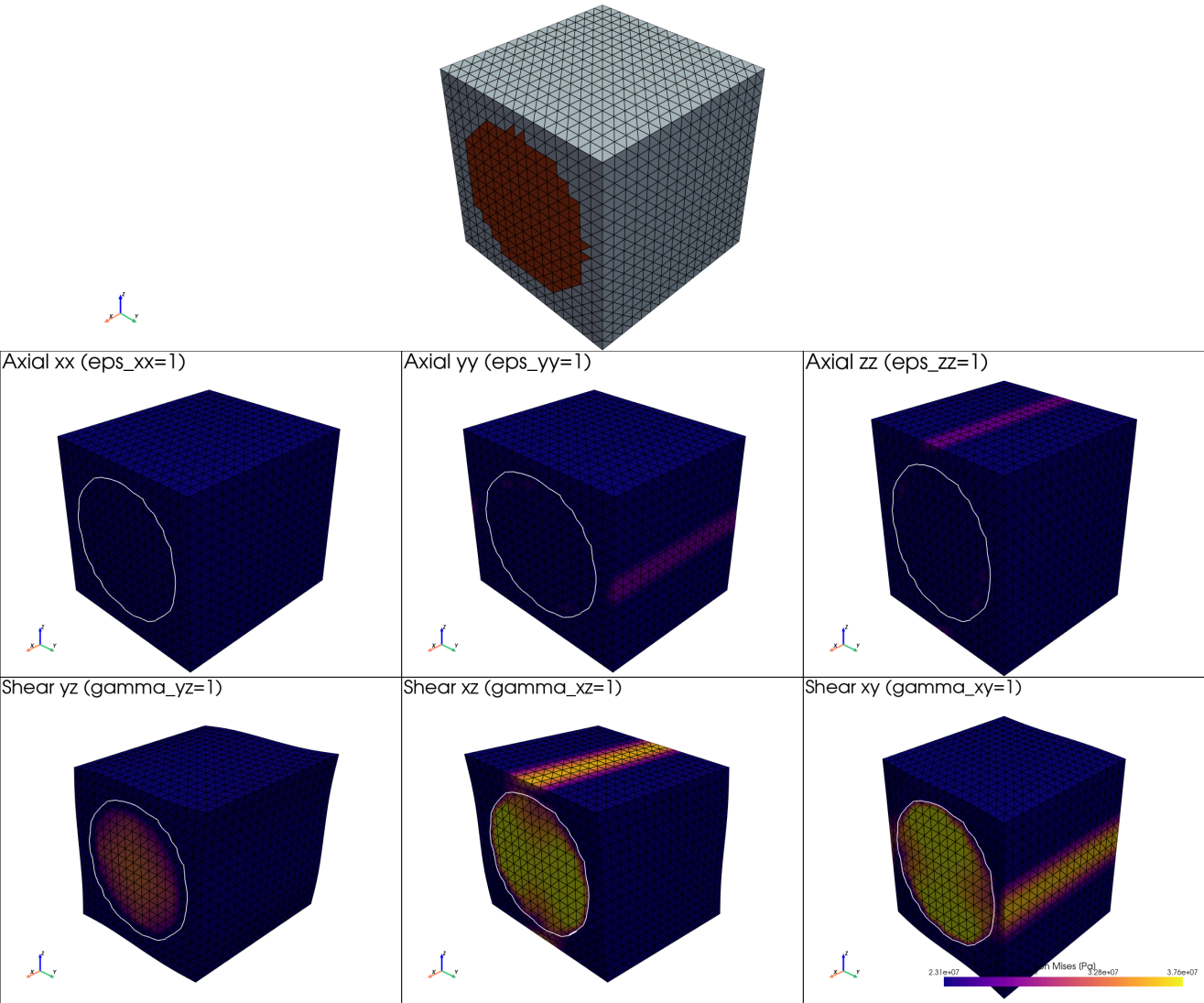


FE mesh (tet) — 16x16x16 structured + 24576 cells; yarn = red, matrix = blue (cell-centroid phase classification)



INPUTS

| material / geometry | values                                                             |
|---------------------|--------------------------------------------------------------------|
| matrix              | E_L=3.0 GPa, E_T=3.0 GPa, G_LT=1.11 GPa, nu_LT=0.35, nu_TT=0.35    |
| yarn                | E_L=140.0 GPa, E_T=10.0 GPa, G_LT=5.00 GPa, nu_LT=0.28, nu_TT=0.40 |
| yarn cylinder       | r=0.4, vf=0.503, axis=(1.0,0.0,0.0)                                |
| mesh                | (16, 16, 16) (tetrahedron)                                         |

OUTPUTS (effective stiffness)

| effective constant     | value               |
|------------------------|---------------------|
| E_x, E_y, E_z [GPa]    | 72.0, 5.9, 5.9      |
| nu_xy, nu_xz, nu_yz    | 0.311, 0.311, 0.435 |
| G_xy, G_xz, G_yz [GPa] | 2.26, 2.26, 1.79    |
| max diag [GPa]         | 74.13               |

FULL C\_eff [GPa] (Voigt order 11,22,33,23,13,12)

|    | 11    | 22    | 33    | 23    | 13    | 12    |
|----|-------|-------|-------|-------|-------|-------|
| 11 | 74.13 | 3.35  | 3.35  | 0.00  | 0.00  | -0.00 |
| 22 | 3.35  | 7.45  | 3.32  | 0.00  | 0.00  | -0.00 |
| 33 | 3.35  | 3.32  | 7.45  | 0.00  | -0.00 | -0.00 |
| 23 | 0.00  | 0.00  | 0.00  | 1.79  | -0.00 | -0.00 |
| 13 | 0.00  | 0.00  | -0.00 | -0.00 | 2.26  | 0.02  |
| 12 | -0.00 | -0.00 | -0.00 | -0.00 | 0.02  | 2.26  |

BOUNDARY CONDITIONS

| where                                                  | boundary condition                                                                                                                                                                                                                                                         |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| this figure (visualization)                            | Periodic: $u(x) = \text{eps\_total} * E_k \cdot x + u\_tilde(x)$ , $u\_tilde$ periodic on opposite faces, non-overlapping slave masks per axis (axis 0 excludes $y=L, z=L$ sub-edges; axis 1 excludes $z=L$ ; axis 2 takes rest), 3 sub-space pins at the geometric centre |
| homogenization solver (b3-tex solve –backend periodic) | Same periodic BC, 6 unit Voigt strains; effective stiffness = volume-averaged stress per loadcase                                                                                                                                                                          |
| KUBC alternative (b3-tex solve –backend kubc)          | $u = E_k \cdot x$ on the entire boundary; gives an upper bound for $C\_eff$                                                                                                                                                                                                |

PERIODIC-BC VERIFICATION (machine-precision residuals over all 6 loadcases)

| check                      | worst residual (across all 6 loadcases) |
|----------------------------|-----------------------------------------|
| u_tilde slave-master pairs | 1.99e-30                                |
| u_tilde 8 corners agree    | 1.79e-30                                |
| u_tilde at centre pin      | 0.00e+00                                |

this looks correct, is this node by node or interpolated?